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MRI Images Based Brain Tumor Detection Using CNN for Multiclass Classification

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***Abstract*— Brain Tumor has ranked as the 10th most common cause of death in recent years. An early brain tumor diagnosis, detect the faster treatment which improves survival of patients. A biopsy is a medical procedure that helps to use brain tumor were diagnosed and treated. The Artificial Intelligence supports the radiologist in tumor diagnosis without invasive measure. Specifically, algorithms based deep learning CNN techniques are used for the detection of brain tumor. In this paper, we proposed 9 layers convolution neural network for identifying and categorizing tumor on the basis of texture, location and shape. The CNN models are trained and tested using the Kaggle (Brain Tumor Classification - MRI) dataset. The experimental result shows that improved accuracy and can be used to assist the decision making to doctor.**

Keywords*— *Brain Tumor Detection, Convolutional Neural Network (CNN), Deep Learning, Magnetic Resonance Imaging (MRI).

I. INTRODUCTION

The human body's most complicated organ is the brain which made up of an enormous number of cells. On average, the human brain contains about 100 billion neurons and approximately 1,000 trillion synapses. A brain tumor is the development of abnormal cells in the human brain [1]. International Association of Cancer Registries (IARC) reported 28,000 brain tumor cases in India every year. Unfortunately, 86% of the cases reported as death. In 2023, there will be 18,990 deaths from primary brain and CNS tumors in the United States.

Brain and nervous system tumors are a complex medical issue with over 130 different types. These tumors can be either benign or malignant. Their growth can cause an increase in pressure inside the skull, leading to brain damage or even death. Primary brain tumor develops in the brain and are frequently benign, whereas secondary brain tumor develops when cancer cells spread from other organs. A meningioma is a tumor that appear in the meninges, and a glioma is a tumor that present in the brain or spinal cord. The pituitary tumor is caused by the pituitary gland's abnormal cell of growth [3].

To determine if a tumor is benign or malignant, biopsies commonly used to identify, but brain tumor often require surgery before definitive diagnosis can be obtained. Therefore, developing an accurate equipment for classification and detection from MRI brain images is essential for precise diagnostics and to minimize the need for surgery. In most cases, oncologists use medical imaging technology like a Computed Tomography (CT) scans and Magnetic Resonance Imaging (MRI) [6]. A high-resolution image of the brain can be obtained with these modalities to observed the changes. We used Kaggle MRI-collected dataset of grayscale [14] images which contains various types of tumor and healthy brain images.

The advancement in the Artificial Intelligence (AI) and imaging modalities gives rise to Computer-Aided Diagnostic (CAD) systems [6] this can be utilized for improving the precision of tumor detection and making better informed decisions. Recently, For the detection and classification of tumor, artificial intelligence approaches including Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Convolutional Neural Networks (CNN). Here we used 9 layers architecture of CNN to increase the accuracy of the brain tumor [12]. The CNN architecture layers are convolutional layers, pooling layers, fully connected layers and output layers. These layers are stacked in a sequential order to form a deep neural network capable of extracting features from images and making predictions [11]. This sequential design feature extraction and classification procedures are combined in CNN's. The other advantage of this architecture is it works on small amount of training data which are more efficient. The main benefit of using a CNN model is that, it makes use of topological information already present in the input to produce highly accurate recognition of brain tumor [15].

II. LITERATURE SURVEY

This literature survey will explore the latest research in CNN using brain tumor detection, along with other relevant reference papers. Many studies have been done to find out how well CNN-based models work for detecting and classifying brain tumor. CNN has emerged as a promising tool for detecting brain tumors accurately and automatically. The use of CNNs in brain tumor detection and diagnosis has the potential to significantly improve patient diagnosis and treatment planning. The finding and categorization of brain tumor is an important challenge due to its presence.

To identify brain abnormalities and to classify as per tumor grades Md. Saikat Islam Khan et. al [01], Introduces two deep learning models "23 layers CNN architecture with VGG16". These proposed models are suitable for large volume of data as well as limited amount of data. Whenever limited data is available the model is fine-tuned with a compressive data augmentation method to improve the model's effectiveness. Francisco Javier Díaz-Pernas et. al [02], MRI scans that have been T1-weighted and contrast-enhanced to test its performance. Elastic data transformation was employed to expand the training dataset and avoid overfitting. Yakub Bhanothu et. al [03], The appropriate bounding box formed by RPN to efficiently identify the

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brain tumor locations. They improved mAP for detecting brain tumor using the test set.

Zarin Anjuman Sejuti et. al [4], The convolutional 2D layers, max-pooling layers, fully connected layers, and batch-normalization layers make up the proposed convolutional neural network, which has 19 layers. Functions of activation ReLU is utilized mostly in CNN-SVM based technique to more precisely classify tumor with a SoftMax classification. Sunil Kumar et. al [5], The classifier performances in terms of accuracy, validation testing and sensitive emotion. Tonmoy Hossain et. al [8], We evaluated six classic classifiers: K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Multilayer Perceptron (MLP), Naive Bayes, Logistic Regression and Random Forest were all of built using the scikit-learn framework, because it performs better than the conventional ones. Then, we proceeded to Convolutional Neural Networks (CNN), which are assembled with TensorFlow and Keras respectively. Traditional classifiers and convolutional neural networks were employing prior to the fuzzy C-Means clustering approach. Milica M. Badža et. al [10], Data testing using record-wise 10-fold cross-validation using confusion matrices.

Hossam H. et. al [12], The proposed network as part of artificial intelligence, ML algorithms are growing increasingly in the field of medical imaging. Using a customized deep neural network, a CAD system classifies brain MRI scans for brain tumor into different classifications (Grades II, III, and IV). The CNN structure consists of 16 layers. Suhib Irsheidat et. al [14], The ACCN will classify MRI images as positive (label equal 1, indicating the existence of a tumor) or negative (label equal 0, indicating the absence of a tumor). An ACNN needs parameters to improve execution and performance, prevent losing control of the model, and keep the ACNN's ability to be accurate. Pallavi Tiwari et. Al [15], Suggested model is tuned using the 32-batch-size parameters of the Adam optimizer, 30-epochs-of-training, categorial cross entropy for the losses, and accuracy as the measure of choice.

TABLE I. LITERATURE SURVEY

Author, Year	Dataset Used	Number of Images	Technique	Layers	Classification Type	Accuracy (%)
Md. Saikat Islam Khan et. al, 2022 [01]	Harvard Medical, Figshare	3064, 152	CNN, VGG16	CNN -23 Layers	Multi Class	Harvard Medical 97.8% and Figshare 100%
Francisco Javier Díaz-Pernas et. al, 2021 [02]	MRI image Dataset	3064	CNN	-	Multi Class	97.3%
Zarin Anjuman Sejuti et. al, 2021 [4]	Figshare	3064	CNN, SVM	CNN -19 Layers	Multi Class	97.1%.
Sunil Kumar et. al, 2021 [5]	Br35H 2020, Central research UK	3060	CNN	-	-	92%
Tonmoy Hossain et. al, 2019 [8]	BRATS	102	CNN, SVM	CNN – 05 layer	Binary Class	CNN 97.87% and SVM 92.42%.
Hossam H. et. al, 2019 [12]	Nanfang Hospital and, Tianjing Medical University	3064, 516	CNN	CNN -16 Layers	Multi Class	Nanfang Hospital 96.13% and Tianjing Medical University 98.7%
Suhib Irsheidat et. al, 2020 [14]	Kaggle	253	CNN	-	Multi Class	88.25%
Pallavi Tiwari et. al, 2022 [15]	Kaggle	-	CNN	-	Multi Class	99%

III. THE PROPOSED SYSTEM

A deep learning-based convolutional neural network (CNN) technique, that used to designed detect and classify brain tumor. We use a dataset of brain MRI images that includes glioma, meningioma, pituitary, and non-tumor images. There will be a training set and a testing set for this dataset. In the training set, we will train a model and then use parameters to assess its outcome on the testing set such as accuracy, precision, sensitivity, specificity, support, F1-score, recall and loss. The finally, model can distinguish input MRI brain images on their respective types. The system enhances brain tumor diagnosis accuracy and speed, resulting in improved patient outcomes. Fig. 1 shown the block diagram of proposed system with its numerous stages.

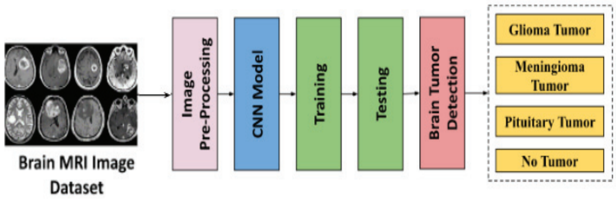

Fig. 1. Block Diagram of Brain Tumor Detection

A. Dataset

We used the publicly available Brain Tumor Detection and Classification MRI image Kaggle dataset are (https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri) [9]. There are 3264 brain MRI images in dataset which have been labelled as glioma tumor, meningioma tumor, no tumor and pituitary tumor. Including the number of these brain MRI images are 926 images of glioma, 937 images of meningioma, 500 images of no tumor and 901 images of pituitary tumor. The images were processed, they were divided into a training and validation set using an 80%-20% split respectively. Depending on their types and grades, brain tumor can vary in terms of size, location and shape. The dataset contains three distinct perspectives: axial, coronal and sagittal views, as illustrated in Fig.2.

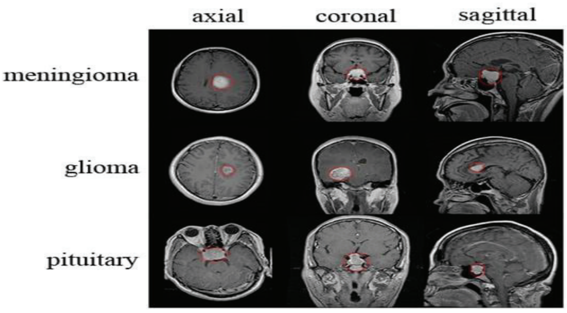

Fig. 2. Different MRI Images of Brain Tumor

B. Image Pre-Processing

The image analysis pre-processing stage using a CNN model resize the input images to a smaller size before feeding them into the model. This resizing is important for increasing the speed of the training process and ensuring that the model can accurately test on new images. For MRI images, the resizing process typically involves reducing the size of the original images from 256 x 256 pixels to a smaller size of 150 x 150 pixels [10].

C. CNN Model

As a subset of deep learning algorithms, CNN (Convolutional Neural Network) models are commonly employed in the fields of image processing and recognition. It is created to automatically identify patterns and features in an image by processing it through a series of layers. CNN designs include AlexNet, GoogleNet, and LeNet, each of which has its unique structure and set of parameters that contribute to its performance [11]. CNN models are a preferred option due to its ability to automatically learn and extract characteristics from images.

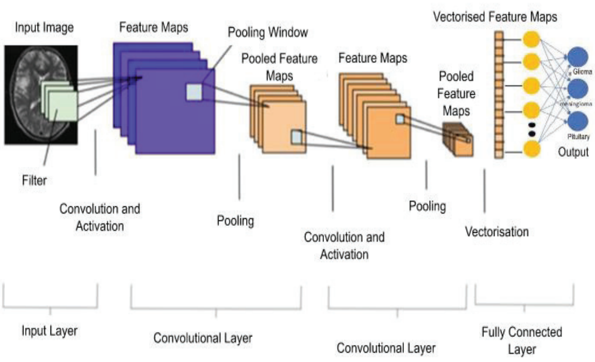

Fig. 3. CNN Architecture

Fig. 3 represents that, the CNN Architecture consists of numerous layers, each with a function. The design of each layer is as follows:

1) Input Layer: Input data, such as an MRI picture, passes through a Convolutional Neural Network (CNN) at the first layer, called the input layer [12]. This layer comprises an array of neurons, where each neuron corresponds to a single pixel in the input image. The pixels of the input image define the dimension of the input layer. The image data represented as a three-dimensional matrix. Each input image has the shape (150, 150, 3).

2) Convolution Layer: The convolutional layer is responsible for performing mathematical operations and applying filters to input images to extract important features. The filters, also known as kernels or weights that are small matrices which assembled with the input picture to generate a feature map to highlights important patterns and structures. The filter size and the values within the filter matrix are learned by the network during training to optimize the performance of the network. Each convolutional layer in the CNN has multiple filters, and thus the output of each filter produces a distinct feature map that is then fed into the subsequent layers for further processing [15].

3) Activation Function: A non-linear activation function used in Convolutional Neural Network (CNNs) is the Rectified Linear Unit (ReLU), which is applied to the output of each neuron in a layer. It is non-linear because its output is not a linear function of its input. The output of the ReLU is identical to the input when it is positive, and it is zero when it is negative. This creates a non-linear output that helps the network learn complex relationships between the input and output, making ReLU a powerful and effective activation function for CNNs [13]. Well-known activation functions used in deep learning models are ReLU, SoftMax and sigmoid Function. [15].

4) Pooling Layer : In general, layer pooling reduces the total amount of numbers and dimensions across layers, which results in smaller tensors than previously. This layer is special and efficient because it reduces the dimensions of the feature map while maintaining the original region feature. There are various pooling techniques, of which using MAX pooling, we extracted the greatest value in a specific dimension [14].

5) Fully Connected Layer : Full connectivity layer often occurs in the last layers of a network. In most cases, the this layer will serve as a direct connection between two adjacent levels.

6) Output Layer: The purpose of output layer in this network is to produce the final output predictions for given input images. It typically consists of one or more neurons, each corresponding to a specific class label or regression target. During training, the output layer learns to map the input features to the correct output class or value and during testing, it produces the final predictions based on the learned mapping.

IV. RESULTS ANALYSIS

This section discusses the experimental results that illustrate the efficacy of the method proposed.

A. Experiment Setup

In this work, Google Colab was used because of its free online cloud service and 15 GB of free space available in Google Drive. Based on Jupyter Notebooks, this cloud service offers a virtual GPU with 12 GB of RAM that is powered by an NVIDIA Tesla K80. This model implements the TensorFlow framework with the Keras library in Python

B. Graph

The dataset of Magnetic Resonance Images (MRI) images are divided into training and testing images. Fig. 4 represents the graph of accuracy. The model was trained using 50 numbers of epochs and its highest accuracy are 88.00%.

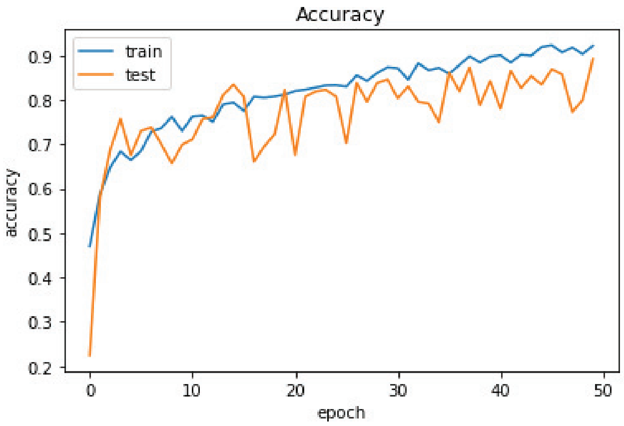

Fig. 4. Accuracy Graph

Fig. 5 Graph displays the model loss values for the CNN model trained on the Magnetic Resonance Images (MRI) dataset over the 50 epochs. The loss is about 0.2% indicate the error rate of the model with lower values indicating better performance.

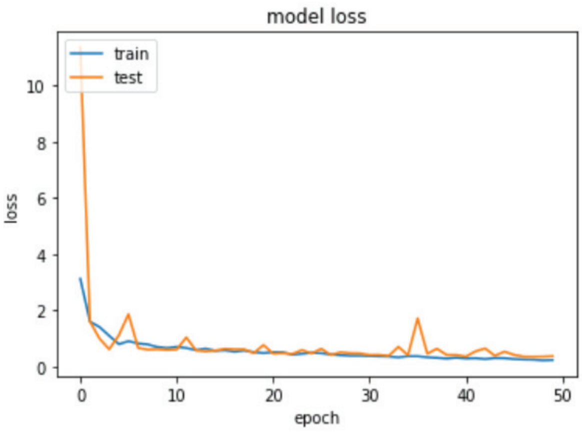

Fig. 5. Model Loss Graph

C. Confusion Matrix

According to the classification of glioma tumor, meningioma tumor, no tumor and pituitary tumor, Fig. 6 displays the CNN model's confusion matrix. In this matrix displays the number of samples that have been correctly classified on the diagonal, and the off-diagonal cells indicate the number of misclassified samples. By analyzing the confusion matrix, we can identify the areas where the model performs well and where it may need further improvements to enhance the classification accuracy.

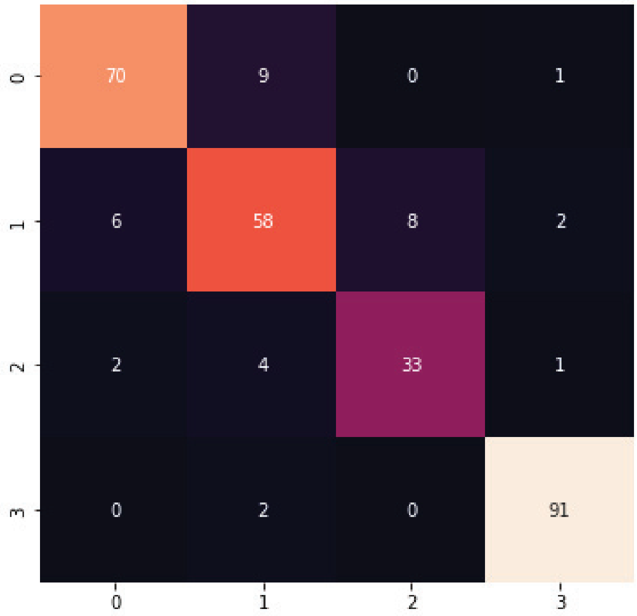

Fig. 6. Confusion Matrix

Following are mathematical equations used to determine the outcome of a classifier according to the evaluation of the confusion matrix.

$$Precision = \frac{TP}{(TP + FP)} \tag{1}$$

$$Sensitivity = \frac{TP}{(TP + FN)} \tag{2}$$

$$Specificity = \frac{TN}{(TN + FN)} \tag{3}$$

$$Accuracy = \frac{TP + TN}{(P + N)} \tag{4}$$

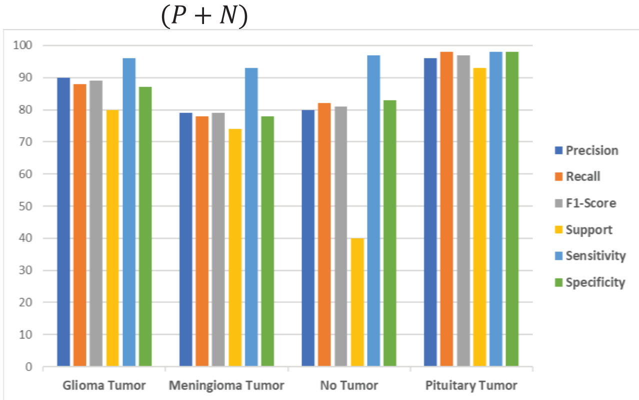

Fig. 7. CNN Model Performance

On the basis of predetermined equations, Fig. 7 shows performance graph of the CNN models.

V. CONCLUSION

In this study, brain tumor is a substantial cause of mortality, and early diagnosis is crucial for improved patient outcomes. Biopsy is an invasive procedure that aids in diagnosis, but recent advancements in Artificial Intelligence have allowed for non-invasive methods for tumor detection and classification. The Convolutional Neural Networks are being utilized to classify brain tumors based on texture, location and shape. The dataset contains a variety of tumor types, including Glioma Tumor, Meningioma Tumor, No Tumor and Pituitary Tumor. The result of test demonstrate accuracy of 88% for the CNN model can assist doctors in decision-making and aid in the timely diagnosis and treatment of brain tumors.

The future scope for the detection of brain tumor through deep learning is promising, as this technology can increase accuracy and decreases time required for the diagnosis. The development of more advanced algorithms and the availability of larger datasets, deep learning can enable better prediction of tumor leading to more effective treatments and better overall healthcare.

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